

CariSurg MedTech Pathways - Week 0

Day 6: Digital Triage Pseudocode

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The design is grounded in the National Early Warning Score 2 (NEW2) framework, which was updated by the Royal College of Physicians in 2017 and has since been mandated across NHS England as the standard early warning system for identifying acutely ill patients, including those with sepsis (Royal College of Physicians [RCP], 2019). Machine Learning (ML) research has shown that models trained on the triage data can achieve strong predictive performance for hospital admission outcomes (Tschoellitsch et al., 2023), but for a first digital triage tool, a rule-based scoring approach grounded in validated clinical thresholds is the safer and more explainable starting point.

First, Inputs, the system receives one row of data per patient. Based on the dataset given and used throughout Week 0, which can be used as an example, the required input columns are:

REQUIRED INPUTS per patient:

Age	(integer, years)
Gender	(binary: 0 = Female, 1 = Male)
GCS	(integer, 3 - 15)
SBP	(float, mmHg)
DBP	(float, mmHg)
MAP	(float, mmHg)
pulse	(float, bpm)
Temp	(float, degrees Celsius)

RR (float, breaths per minute)

Fio2 (float, percentage - 21 = room air, >21 = supplemental O2)

Second, data validation and cleaning, before any scoring can happen, the system must validate and clean each input. Raw clinical data missing values, text variants and physiologically impossible entries, exactly the problems identified in days 1 and 2 of the work done in week 0. A digital triage tool that does not handle dirty data will produce dangerously wrong risk scores (Dewitte et al., 2022)

Third, NEWS2 Scoring, once the data is clean, the system scores each vital sign using the official NEWS2 point table. NEWS2 allocates 0-3 points per parameter based on how far the value deviates from normal, then sums them into a total score (RCP, 2017). An additional 2 points are added if the patient is on supplemental oxygen.

Fourth, SIRS Sepsis Flag, NEWS2 alone does not catch all sepsis presentations. A parallel SIRS check is run as a secondary safety layer. Research has shown that combining NEWS2 with the SIRS criteria improves sepsis sensitivity at triage (Dewitte et al., 2022).

Fifth, Composite shock flag, separately from NEWS2, the system checks for haemodynamic compromise using the shock index pattern identified in day 2 of the portfolio.

Sixth, Risk Categorization, the system combines the NEWS2 total score with the SIRS and shock flags to assign a final risk level. The NEWS2 risk thresholds are taken directly from the RCP (2017) framework. Any positive safety flag automatically escalates the risk level upward, even if the NEWS2 score alone would suggest a lower category.

Seventh, Clinical Response trigger, each risk category maps to a defined clinical response, following the NEWS2 escalation framework (RCP, 2017). The clinical response protocol is determined by the NEWS2 score, which categorizes patient risk into four distinct levels. A Low Risk patient with a score of 0 to 4 requires routine monitoring and scheduled

reassessments, whereas a Low-Medium Risk is triggered by a "Red Flag." This flag represents any single parameter scoring 3 points, such as a GCS below 13, systolic blood pressure below 90, or a respiratory rate of 25 or higher, and it necessitates increased monitoring and notification of the nurse in charge. For patients at Medium Risk with a score of 5 to 6, an urgent review by a doctor or acute care team is required. Finally, a High Risk patient with a score of 7 or greater demands an immediate review and consideration for ICU escalation. If a shock or SIRS flag is present at this high risk level, the sepsis bundle is automatically activated.

Full System flow:

FOR EACH patient arriving at triage:

1. RECEIVE raw data row from EHR or nurse tablet entry
2. CALL `validate_and_clean(patient_row)`
→ returns `cleaned_row`, `missing_flag`
3. IF `missing_flag == True`:
LOG missing fields for clinical review
CONTINUE with imputed values but mark output as INCOMPLETE
4. CALL `calculate_news2(cleaned_row)` → returns `news2_score`
5. CALL `check_sirs(cleaned_row)` → returns `sepsis_flag`, `sirs_count`
6. CALL `check_shock(cleaned_row)` → returns `shock_flag`

7. CALL assign_risk_level(news2_score, sepsis_flag, shock_flag → returns
final_category, reason

8. OUTPUT to triage dashboard:

Patient ID : [ID]

Age / Gender : [Age] / [Gender]

NEWS2 Score : [news2_score]

SIRS Criteria : [sirs_count] / 3 available

Shock Flag : [Yes / No]

RISK LEVEL : [final_category]

Reason : [reason]

Data Complete : [Yes / No]

9. IF final_category == "HIGH":

SEND immediate alert to nurse in charge

PRINT bed assignment request

END

Lastly, limitations and design notes, This system serves strictly as a clinical decision support tool, meaning all outputs require human oversight and cannot replace a nurse's direct assessment (Tschoellitsch et al., 2023). Its performance relies heavily on data quality: missing vitals are imputed using population medians, which lowers prediction confidence and triggers a missing_flag to warn clinicians (Miles et al., 2020). To mitigate the known limitations and low sensitivity of standalone SIRS screening (Dewitte et al., 2022), SIRS is integrated as a secondary layer to supplement NEWS2 and guide clinical judgment (RCP, 2017).

Furthermore, the current dataset lacks critical variables like SpO₂, white blood cell count and chief complaint, all of which must be collected at triage for a production-ready version to ensure accurate sepsis detection.

References

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